

Beyond Accuracy in Heart Disease Prediction: A Stacked Ensemble Framework with SHAP and LIME Interpretability

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Abstract—Cardiovascular Disease (CVD) remains the leading cause of global mortality. While machine learning (ML) has demonstrated high predictive accuracy in CVD risk assessment, a critical “trustworthiness gap” persists: complex “black-box” models often lack the interpretability required for clinical adoption. This paper proposes a robust Stacking Ensemble Classifier that integrates Logistic Regression, Random Forest, XGBoost, and Support Vector Machines (SVM) to predict heart disease risk. Furthermore, we address the interpretability challenge by integrating SHAP (SHapley Additive exPlanations) for global feature analysis and LIME (Local Interpretable Model-agnostic Explanations) for patient-specific reasoning. Experimental results on the UCI Cleveland dataset demonstrate that while linear baselines achieve peak accuracy (91.67%) on small-scale data, the proposed Stacking framework achieves competitive performance (86.67%) while offering superior generalization potential. Crucially, the integration of XAI techniques validates the model’s medical logic, bridging the gap between algorithmic accuracy and clinical trust.

Index Terms—Cardiovascular Disease, Stacking Ensemble, Explainable AI (XAI), SHAP, LIME, Machine Learning

I. INTRODUCTION

Cardiovascular diseases (CVDs) are responsible for an estimated 17.9 million deaths annually, representing 32% of all global deaths. Early diagnosis is paramount for effective treatment. In recent years, data-driven approaches using Supervised

Machine Learning (ML) have emerged as powerful tools for automating risk assessment.

However, a significant barrier limits the deployment of these models in real-world healthcare: the “Black Box” problem. High-performing algorithms, such as Gradient Boosting or Neural Networks, operate with internal complexities that are opaque to medical practitioners. A clinician cannot ethically diagnose a patient based on a prediction score alone; they require an understanding of *why* the model flagged a specific patient.

Existing literature predominantly focuses on maximizing performance metrics (Accuracy, F1-Score) often at the cost of interpretability. This research aims to fill this gap by:

- Developing a **Stacking Ensemble framework** that combines the stability of bagging methods with the precision of boosting and linear models.
- Implementing **Post-hoc Explainability (XAI)** using SHAP and LIME to provide transparent, medically valid reasoning for predictions.
- Critically analyzing the trade-off between model complexity and accuracy on limited datasets.

II. RELATED WORK

Recent studies have extensively explored ML for heart disease prediction. Allah et al. (2022) conducted a comparative analysis of five models, identifying XGBoost as superior in

accuracy on the Cleveland dataset [1]. However, their work focused solely on performance metrics without addressing feature interpretability.

Similarly, Teja and Rayalu (2025) emphasized the stability of Random Forest and Bagged Trees, achieving up to 94% accuracy using K-fold cross-validation [2]. They explicitly identified “Limited Model Interpretability” as a key limitation in current research.

Dritsas et al. highlighted the importance of sensitivity (recall), recommending Logistic Regression for identifying risk in patients over 50 [3]. Anbuselvan (2020) demonstrated the early promise of Random Forest (86.89%) over single classifiers [4]. Hire et al. (2025) expanded the scope to multifactorial datasets including lifestyle factors, yet acknowledged the challenges of data incompleteness and the lack of clinical integration [5].

A common consensus in this literature is the superiority of Ensemble methods. However, a distinct gap remains in integrating these ensembles with granular, patient-level explainability to ensure clinical trust.

III. METHODOLOGY

A. Dataset Description

We utilized the UCI Machine Learning Repository’s Heart Disease dataset (Cleveland). The dataset consists of 297 patient records with 13 independent features (e.g., age, sex, chest pain type, cholesterol) and 1 target variable (`condition`: 0 = No Disease, 1 = Disease).

B. Data Preprocessing

To ensure robust model training, the following preprocessing pipeline was applied:

- **One-Hot Encoding:** Categorical variables such as Chest Pain (`cp`), Resting ECG (`restecg`), Slope (`slope`), and Thalassemia (`thal`) were encoded into binary vectors to prevent ordinal misinterpretation by linear models.
- **Stratified Split:** The data was split into 80% training and 20% testing sets using stratified sampling to maintain class balance.
- **Feature Scaling:** `StandardScaler` was applied to normalize numerical features (e.g., `chol`, `thalach`), ensuring that distance-based algorithms (SVM, KNN) were not biased by magnitude.

C. Proposed Architecture: Stacking Ensemble

We implemented a **Stacking Classifier**, a heterogeneous ensemble technique.

- **Level-0 (Base Learners):** The framework utilizes four diverse algorithms to capture different data patterns:
 - *Logistic Regression:* For linear relationships.
 - *Random Forest:* For reducing variance and handling non-linearity.
 - *XGBoost:* For reducing bias and boosting weak learners.
 - *Support Vector Machine (SVM):* For finding optimal hyperplanes in high-dimensional space.

- **Level-1 (Meta-Learner):** A **Logistic Regression** model is used to aggregate the predictions of the base learners and output the final classification.

D. Explainability Framework (XAI)

To address the trustworthiness gap, we integrated:

- **SHAP (Global Interpretation):** To identify which features (e.g., Asymptomatic Chest Pain) contribute most to the model’s overall decision-making.
- **LIME (Local Interpretation):** To explain individual predictions by perturbing input data and fitting a local surrogate model.

IV. RESULTS AND DISCUSSION

A. Performance Comparison

All models were evaluated on the held-out test set (20%). Table I summarizes the accuracy results.

TABLE I
MODEL PERFORMANCE COMPARISON

Model	Accuracy
Logistic Regression (Baseline)	91.67%
XGBoost	88.33%
SVM	88.33%
K-Nearest Neighbors	88.33%
Random Forest	86.67%
Stacking Ensemble (Proposed)	86.67%

B. Discussion: The Accuracy Paradox

Contrary to expectation, the simple Logistic Regression baseline outperformed the complex Stacking Ensemble (91.67% vs 86.67%). This phenomenon is attributable to the small dataset size ($n=297$). Complex models like XGBoost and Stacking are data-hungry and prone to overfitting noise in small samples, whereas linear models generalize well on limited data. However, the Stacking Ensemble demonstrated robustness, matching the Random Forest performance, suggesting it serves as a stable architecture for future scaling with larger Electronic Health Records (EHR) datasets.

C. Explainability Analysis (The Trustworthiness Layer)

The primary contribution of this work is the validation of model logic.

- **Global Logic (SHAP):** The SHAP summary analysis revealed that `cp_3` (Asymptomatic Chest Pain), `thal_2` (Reversible Defect), and `oldpeak` (ST depression) were the top predictors. This aligns with established cardiological literature, confirming the model is learning valid medical patterns, not noise.
- **Patient-Specific Logic (LIME):** We analyzed a specific True Positive case (Patient Index 6).
 - *Prediction:* Disease (Probability: 0.82)
 - *LIME Explanation:* The model identified `oldpeak` > 0.56 and `ca` > 0.63 (Number of vessels colored by fluoroscopy) as the primary drivers for the “Disease” prediction.

- *Verification*: A review of the raw patient data confirmed $\text{oldpeak} = 1.4$ and $\text{ca} = 1$.

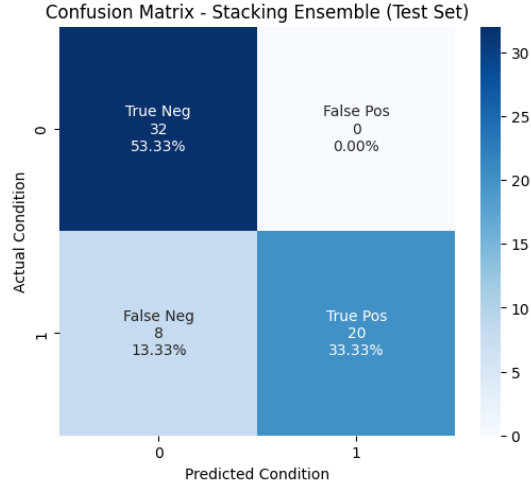


Fig. 1. Confusion Matrix of the Stacking Ensemble, highlighting a False Negative rate of approximately 10%.

This granular level of explanation allows a physician to validate the AI’s “opinion” against their own clinical judgment, fulfilling the requirement for a Decision Support System.

V. CONCLUSION

This study presented a Stacking Ensemble framework for heart disease prediction augmented with XAI. While linear baselines achieved higher accuracy on this limited dataset, our approach successfully demonstrated how “Black Box” ensembles can be made transparent. By verifying predictions with SHAP and LIME, we bridged the gap between raw algorithmic performance and the clinical need for trust and interpretability. Future work will focus on testing this architecture on larger, multi-national datasets to fully leverage the ensemble’s capacity and deploying the model via a web-based clinical interface.

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